

Regression Assignment

Date: 02.10.2026

Problem Statement:

A client's requirement is; he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same. As a data scientist, you must develop a model which will predict the insurance charges.

- 1.) Identify your problem statement
- 2.) Tell basic info about the dataset (Total number of rows, columns)
- 3.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)
- 4.) Develop a good model with r^2 _score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.
- 5.) All the research values (r^2 _score of the models) should be documented. (You can make tabulation or screenshot of the results.)
- 6.) Mention your final model, justify why u have chosen the same.

1) Identify your problem statement

Need to predict the **Insurance Charges** based on the given data using the best Machine Learning model.

Step:1 Machine Learning <---(Both Numerical and Categorical values are present in the dataset)

Step:2 Supervised <---(Input and Output are clearly mentioned)

Step:3 Regression <---(Output value will be numerical ('float'))

2.) Tell basic info about the dataset (Total number of rows, columns)¶

Number of Columns: 1338

Number of Rows: 6

3.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

Data Pre-processing:

Step:1 Converting the nominal data into numerical data

Step:2 Splitting the Input and Output variables

Step:3 Using sklearn.model_selection, splitting the dataset into Training and Test set

4.) Develop a good model with `r2_score`. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model. [1](#)

Machine Learning Algorithms:

1. Multiple Linear Regression
2. Support Vector Machine
3. Decision Tree
4. Random Forest

5.) All the research values (`r2_score` of the models) should be documented. (You can make tabulation or screenshot of the results.)

1. Multiple Linear Regression

Result: `r2 score = 0.71`

<- Poor Score

2. SVM

Iterations	kernal	C	R2_score	Standarization
1	linear	1	-0.01	Yes
2	linear	10	0.462	Yes
3	linear	100	0.629	Yes
4	linear	1000	0.765	Yes
5	linear	2000	0.744	Yes
6	linear	3000	0.741	Yes
7	poly	1000	0.857	Yes
8	poly	2000	0.861	Yes
9	poly	3000	0.86	Yes
10	rbf	1000	0.81	Yes
11	rbf	2000	0.855	Yes
12	rbf	3000	0.866	Yes
13	sigmoid	100	0.528	Yes
14	sigmoid	200	0.546	Yes
15	sigmoid	300	0.5	Yes

3. DECISION TREE

Iterations	criterion	splitter	R2_score	Standarization
1	squared_error	best	0.687	Yes
2	poisson	best	0.736	Yes
3	absolute_error	best	0.653	Yes
4	friedman_mse	best	0.686	Yes
5	squared_error	random	0.726	Yes
6	poisson	random	0.649	Yes
7	absolute_error	random	0.724	Yes
8	friedman_mse	random	0.708	Yes

4. RANDOM FOREST

Iterations	n_estimators	criterion	max_features	R2_score
1	10	squared_error	1	0.851
2	50	squared_error	1	0.858
3	100	squared_error	1	0.855
4	10	absolute_error	1	0.83
5	50	absolute_error	1	0.851
6	100	absolute_error	1	0.849
7	10	poisson	1	0.849

8	50	poisson	1	0.859
9	100	poisson	1	0.857
10	10	squared_error	sqrt	0.854
11	50	squared_error	sqrt	0.872
12	100	squared_error	sqrt	0.874
13	10	absolute_error	sqrt	0.853
14	50	absolute_error	sqrt	0.871
15	100	absolute_error	sqrt	0.871
16	10	poisson	sqrt	0.847
17	50	poisson	sqrt	0.87
18	100	poisson	sqrt	0.872
19	10	squared_error	log2	0.854
20	50	squared_error	log2	0.872
21	100	squared_error	log2	0.874
22	10	absolute_error	log2	0.853
23	50	absolute_error	log2	0.871
24	100	absolute_error	log2	0.871
25	10	poisson	log2	0.847
26	50	poisson	log2	0.87
27	100	poisson	log2	0.872

6.) Mention your final model, justify why u have chosen the same.

Random Forest works well. In which, iteration-12 & 21 are giving the same R2 score as **0.874**. R2 score of **0.874** is a larger value compared to all other models. Since I am selecting this as the best model for deployment.